Hyperspectral Image Denoising via Texture-Preserved Total Variation Regularizer

Yang Chen, Wenfei Cao, Li Pang, Jiangjun Peng, *Graduate Student Member, IEEE*, and Xiangyong Cao

***Abstract*— The total variation (TV) regularizer is a widely used technique in image-processing tasks to model an image's local smoothness property. Intrinsically, the TV regularizer imposes sparsity constraints on the gradient maps of the image, which inevitably weakens the image texture structure and thus affects the quality of image restoration. To alleviate this issue, we propose a novel texture-preserved TV (TPTV) regularizer for hyperspectral images (HSIs) by introducing a weighting scheme. Specifically, the weights are assigned to the gradient maps of HSIs, which help slack the sparsity constraint for the pixels with large variations, thus preserving the texture structure. Additionally, we elaborate an empirical method to learn the weights adaptively from observed HSIs. Then, we propose an HSI denoising method based on the TPTV regularizer. Experimental results on synthetic and real HSIs illustrate the superiority of our proposed method over other state-of-the-art methods. In addition, the proposed weighting scheme can be finely embedded into other TV regularizers and protect the image texture. The experimental results also demonstrate that the denoising performance of the original method is significantly improved after embedding the weighting scheme.**

***Index Terms*— Hyperspectral image (HSI) denoising, texture-preserved total variation (TPTV), total variation (TV), weighting scheme.**

I. Introduction

HYPERSPECTRAL images (HSIs) can be applied in many tasks, for example, terrain detection, military surveillance, mineral exploration, and agriculture because they contain an array of 2-D images collected over continuous wavelengths and supply a wealth of spectral information [1], [2], [3], [4]. Unfortunately, due to the atmospheric interference, photon effects, and physical mechanism, HSIs are inevitably corrupted by various noises during the imaging procedure, for example, Gaussian noise, sparse noise (stripe, deadline, or impulse), and so on [5], [6], [7], [8]. The presence of noise will significantly degrade image quality and negatively impact the performance of the downstream HSI processing tasks, for example, unmixing [9], [10], classification [11], [12], [13], and target detection [14]. Therefore, it is of great importance to denoise the HSIs.

Many methods have been proposed for the HSI denoising problem, among them the deep-learning-based approaches have received extensive attention. For instance, HSID-CNN [15] introduced a deep residual convolutional neural network (CNN) that incorporates multiscale and multilevel feature representation. To enhance the understanding of spatial and spectral correlations, a modified 3-D U-net architecture [16] was presented to leverage the spatial–spectral correlations. Later, the global reasoning network [17] was proposed with three well-designed modules. Furthermore, SQAD [18] and QRNN3D [19] proposed quasi-attention recurrent networks to learn the correlations of spatial features within the spectral domain. More recently, transformer-based approaches have shown promise in HSI denoising, including methods like TRQ3DNet [20], SST [21] and STAS [22]. While these methods have demonstrated outstanding denoising results, they often lack theoretical support and struggle to generalize effectively to new datasets.

In recent decades, advanced model-based methods have demonstrated remarkable success in HSI denoising. These methods explore the inherent prior information of HSIs and explicitly encode them in the denoising model. The most commonly utilized prior knowledge in HSI denoising models includes spatial and spectral correlation, self-similarity, and local smoothness. Specifically, the spatial correlation is due to that there are usually distributed similar objects within a local spatial area, and the spectral correlation is due to the similarity of sensor parameters setting for image acquisition in adjacent spectral bands. To model such prior, the low-rank matrix decomposition and low-rank tensor decomposition models were proposed [23], [24], [25], [26], [27], [28]. The source of self-similarity prior is that similar patterns are usually distributed within local space, and thus similar image patches can be aggregated together, and then low-rank filtering are utilized to remove noise [29], [30], [31], [32]. The local smoothness priors are generated because images are usually composed of some small patterns, such as tables, chairs, books, and so on within these patterns, the pixel values change slightly, such as the surface of a table, showing strong smoothness, while at the edges of these patterns, the pixel values change greatly. Therefore, the pixel values except those located at the edge of patterns are nearly zero in the gradient maps of an image,

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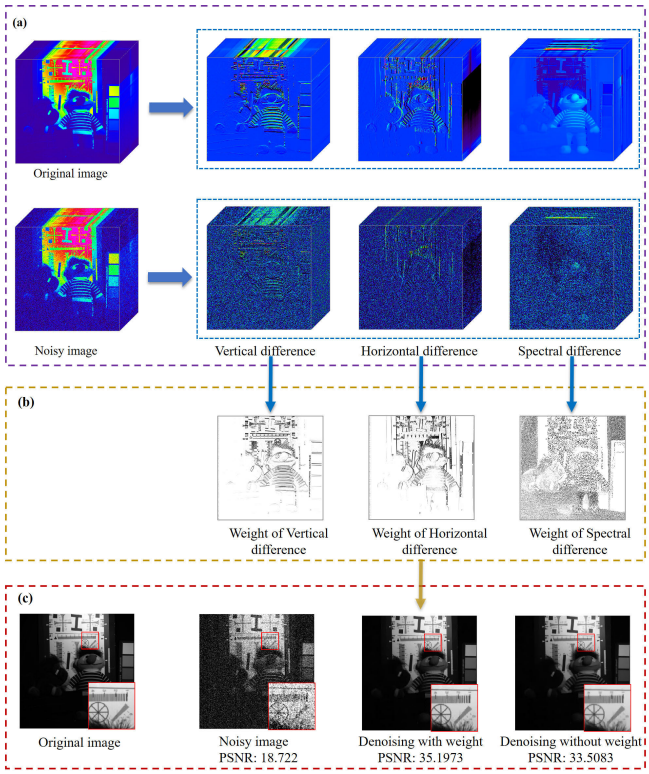


Fig. 1. Illustration of the weighting scheme involved in the proposed TPTV model and the effects on texture preserving. (a) First row: the gradient maps generated from the original HSI among spatial vertical, horizontal, and spectral modes show the image texture among different modes. Second row: the gradient maps generated from noisy HSIs, the SNR is much lower than noisy HSIs. (b) Weight generated from noisy HSIs by the proposed weighting scheme. (c) Denoising performance comparison of the proposed method with or without the weight scheme adopted in the TPTV regularizer.

and thus the sparsity model was adopted to gradient maps that are the total variation (TV) model [33], [34]. In addition, the local smoothness characteristics of HSIs also exist among the spectral domain because the pixel values between adjacent bands usually change gently due to similar sensor parameter settings. Therefore, the advanced spatial–spectral TV (SSTV) models were proposed for HSIs [35], [36], [37], [38].

The TV model imposes the sparsity constraint on gradient maps and was used as a regularization term in the denoising model to help to suppress noise. In most TV-based HSI denoising models, the sparse constraints are applied uniformly to all points in each gradient map, which essentially assume the difference values obey the Laplace distribution independently and identically. This assumption deviates from the reality that pixels in an HSI exhibit different variations, and even the same pixel shows different variations along different directions. For instance, pixels located at pattern edges often have larger variations than those at the pattern surface, as depicted in Fig. 1(a). Consequently, these TV models tend to weaken the contrast at pattern edges when suppressing noise. This is problematic because edges represent the texture structure of the image, which is crucial for capturing structural information. The quality of the image will be greatly damaged if the texture structure is smoothed. To address this issue, we propose a novel texture-preserved TV (TPTV) regularizer by using the Laplacian scale mixture (LSM) distribution model. This approach introduces a new weighting scheme in the TV model for HSIs to protect the texture information. Specifically, we assume that the difference values follow the LSM distribution, and we employ the expectation maximum algorithm (EM) to maximize the log-likelihood function. The optimization objective function in the EM algorithm is the weighted l_1 norm of difference values. The weight assigned to each pixel is determined by the estimated gradient maps, which form a 2-D matrix corresponding to the spatial position. Building upon the LSM model, we define the TPTV regularizer as the weighted l_1 norm of gradient maps, that is, the optimization objective function of the EM algorithm. Consequently, utilizing the TPTV regularizer as a regularization term equals maximizing the log-likelihood function.

How to determine the weight according to gradient maps of the image is the key to the proposed TPTV regularizer. To preserve the image texture and prevent excessive smoothing, the pixels with larger variations should be assigned smaller weights to weaken the sparsity constraints of these points. Conversely, pixels with smaller variations should be assigned larger weights to strengthen their sparsity constraints. Therefore, the weights need to align well with the texture information in the image. However, extracting the image texture accurately is challenging due to the presence of noise, as depicted in Fig. 1(a). To alleviate this issue, we employ low-rank filtering on the denoised image during the denoising process to obtain an image, denoted as $\mathcal{X}_0$, with a higher signal-to-noise ratio (SNR). Unfortunately, the noise remaining in the gradient maps of $\mathcal{X}_0$ still adversely affects the texture structure. Analyzing the randomness of noise and consistency of the image structure, we adopt the mean filter along the spectral dimension to further eliminate the effect of noise to get a reliable estimation of the texture information. Subsequently, the resulting texture information is converted into weight based on the principle that the weight should be inversely proportional to the image texture, as shown in Fig. 1(b). The whole process of mapping HSI data to weights is adaptive and efficient. Finally, we propose an HSI denoising method based on the TPTV regularizer. The experiments show that the proposed denoising model is superior to current advanced methods, which indicates that the proposed TPTV regularizer encodes the local smoothness prior of HSIs better than other TV regularizers. Most noticeably, the proposed weighting scheme can be easily incorporated into other TV regularizers. The experimental results validate that this weighting scheme significantly alleviates the tendency of the original TV regularizer to blur the image texture, as shown in Fig. 1(c).

The main contributions of this article can be summarized as follows.

First, we propose a TPTV regularizer by modeling the distribution of difference values with the LSM distribution. The TPTV regularizer introduces a weighting scheme that could relax the sparsity constraints on the pixels with large variations, effectively preserving the texture information and preventing oversmoothing. This weighting scheme can be easily incorporated into other TV regularizers, enhancing their ability to protect image texture.

Second, we propose a new HSI denoising model called ETPTV by utilizing the proposed TPTV regularizer. To solve

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this model efficiently, we design an efficient alternating direction method of multipliers (ADMM) algorithm.

Third, the experimental results confirm that the proposed denoising method can perform better than other advanced approaches. In addition, the transferring experiments show that the denoising performances can be obviously improved after embedding our weighting scheme into the existing TV regularizers.

The rest of this article is organized as follows. Section II reviews the related works. The notations and preliminaries are presented in Section III. Section IV introduces the weighting scheme, the TPTV regularizer, and the ETPTV denoising model. Section V reports the experimental results. Section VI provides the conclusion.

II. Related Work

The TV regularizers have been widely used in imaging processing tasks, for example, restoration [33] and super-resolution [39]. Very recently, TV regularizers have also been applied to the HSI denoising task. One common way is applying the traditional TV model to HSI band by band and then adding them together (HTV), such as LRTV [40]. However, the HTV regularizer ignores the variation in noise intensity and image scale, which limits its performance.

Naturally, TV regularizers that take into account the characteristics of HSIs have been attracting more attention. Zhang [35] first proposed the cubic TV regularizer, which consists of a spatial 2-D TV model and a spectral 1-D TV model. Considering that noise intensity varies from pixel to pixel, Yuan et al. [36] proposed a spectral–spatial adaptive TV regularizer (SSAHTV) which is capable of automatically adjusting the denoising strength of each pixel. Later, Jiang et al. [37] improved the setting of weights in SSAHTV by considering the positive relationship between the weights and the noise intensity as well as the negative correlation between the weights and the variation of the pixel. Although these two regularizers have some image texture protection effect, they coupled two spatial gradient maps together. Wang et al. [41] explored the significant distinction of gradient map distribution, and it is better to model them separately. Then, Chen et al. [42] proposed an adaptive SSTV (ASSTV) regularizer by independently calculating the weight for each gradient map. Moreover, Peng et al. [34] established a TV regularizer on the spatial representation coefficient of HSIs, thus obviously reducing the running time. Aggarwal and Majumdar [43] designed a novel SSTV regularizer by using the traditional TV model on the gradient map along the spectral mode. This regularizer has been applied to follow-up HSI denoising methods [44], [45]. Furthermore, exploring the local smoothness characteristics of the HSI, Peng et al. [38], [46] found that the spectral correlation of the original HSI also existed in its gradient maps, and then they applied the sparsity constraint to the spatial representation coefficient of each gradient map. Chen et al. [47] found the shared group sparsity pattern in the gradient maps and then proposed a new TV model with weighted $l_{2,1}$ norm. Later, Xue et al. [48], [49] encoded the nonlocal smoothness with a reweighted l_1 minimization scheme and maximum a posterior (MAP) scheme under LSM distribution assumption, respectively. These TV regularizers have demonstrated significant advantages in HSI processing tasks.

III. Notations and Preliminaries

We represent the tensor with decorated letters $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ and its elements are denoted by $\mathcal{X}_{(i_1,i_2,\ldots,i_N)}$. Moreover, the matrices, vectors, and scalars are denoted with nonbold upper case letters X, bold lower case letters $\boldsymbol{x}$, and nonbold letters x.

An HSI is treated as a tensor $\mathcal{X} \in \mathbb{R}^{M \times N \times B}$, where M is the spatial height, N is the spatial width, and B denotes the spectrum number. The HSI tensor $\mathcal{X}$ can be unfolded as a matrix $X \in \mathbb{R}^{MN \times B}$ along the spectral mode by vectorizing the image of each band and then arranging the vectors into a matrix along the order of the spectrum. Based on the spectral correlation characteristic, the HSI can be assumed generated from a low-dimensional spectral subspace. Therefore, the HSI tensor $\mathcal{X}$ can be modeled by the following low-rank decomposition model:

$$\mathcal{X} = \mathcal{U} \times_3 V^T \tag{1}$$

where $\mathcal{U} \in \mathbb{R}^{M \times N \times r}$ is the representative coefficient, $V \in \mathbb{R}^{B \times r}$ is the spectral basis, and $r (r \leq B)$ is the dimension of the spectral subspace. The element of mode-3 multiplication of the tensor $\mathcal{U}$ with the matrix V is defined as follows:

$$(\mathcal{U} \times_3 V^T)_{(m,n,b)} = \sum_{i=1}^{r} \mathcal{U}_{(m,n,i)} V_{(b,i)}. \tag{2}$$

The gradient map of $\mathcal{X}$ along ith mode is denoted as $\mathcal{G}_i^{\mathcal{X}}$, which is defined as follows:

$$\mathcal{G}_i^{\mathcal{X}} = \boldsymbol{D}_i \mathcal{X}, \quad i = 1, 2, 3. \tag{3}$$

The operators $\boldsymbol{D}_1$, $\boldsymbol{D}_2$, and $\boldsymbol{D}_3$ represent the first-order finite-difference operator along the spatial horizontal, spatial vertical, and spectral modes, respectively.

IV. Proposed Model

In this section, we first propose the TPTV regularizer and provide the technical details, then we propose the ETPTV denoising model based on the TPTV regularizer and finally utilize the ADMM algorithm to solve the denoising model.

A. TPTV Regularizer

For HSIs, many advanced TV regularizers have been proposed, such as traditional 3-D TV regularizer [41], enhanced 3-D TV regularizer (ETV) [38], and correlated TV model (CTV) [34]. Here, we propose a unified paradigm for these TV regularizers as follows:

$$\|\mathcal{Z}\|_{3\text{DTV}} = \|\boldsymbol{D}_1 \mathcal{Z}\|_1 + \|\boldsymbol{D}_2 \mathcal{Z}\|_1 + \|\boldsymbol{D}_3 \mathcal{Z}\|_1 \tag{4}$$

where $\mathcal{Z}$ could be the HSI as traditional 3DTV [41], the coefficient of HSIs as CTV [34], or the coefficient of gradient map of HSIs as ETV [38], [46].

The assumption underlying the 3DTV model is that the difference values obey the Laplace distribution independently and identically. This assumption is too strong and deviates

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from the reality that the variations that essentially existed on pixels vary significantly. Therefore, we propose a more refined assumption that the difference of each pixel independently obeys the LSM distribution [49], [50] which can be expressed as follows:

$$\begin{aligned} p(\mathcal{G}) &= \prod_{i,j,b} p(\mathcal{G}_{(i,j,b)}) \\ p(\mathcal{G}_{(i,j,b)}) &= \int p(\mathcal{G}_{(i,j,b)}|\Theta_{(i,j)})p(\Theta_{(i,j)})d\Theta_{(i,j)} \\ p(\mathcal{G}_{(i,j,b)}|\Theta_{(i,j)}) &= \frac{\Theta_{(i,j)}}{2}\exp(-\Theta_{(i,j)}|\mathcal{G}_{(i,j,b)}|) \end{aligned} \tag{5}$$

where $\mathcal{G}_{(i,j,b)}$ denotes the element in the gradient map $\mathcal{G}$ located at the ith row, the jth column, and the bth band. Here, we make some notational simplifications and do not refer specifically to the direction of the difference. The closed-form expression of $p(\mathcal{G})$ usually does not exist, and thus we adopt the EM algorithm to obtain the maximum of the log-likelihood function $\ln(p(\mathcal{G}))$.

1) E Step: We calculate the evidence lower bound (ELBO) of $\ln(p(\mathcal{G}))$ at current estimated $\mathcal{G}^{(t)}$. The ELBO function is denoted as $Q(\mathcal{G}, \mathcal{G}^{(t)})$ which can be expressed as follows:

$$\begin{aligned} &Q(\mathcal{G}, \mathcal{G}^{(t)}) \\ &= \int \prod_{i,j} p(\Theta_{(i,j)}|\mathcal{G}^{(t)}) \ln \prod_{i,j,b} p(\mathcal{G}_{(i,j,b)}|\Theta_{(i,j)})d\Theta_{(i,j)} \\ &= -\sum_{i,j,b} \langle \Theta_{(i,j)} \rangle |\mathcal{G}_{(i,j,b)}| + \text{const.} \end{aligned} \tag{6}$$

where $\langle \cdot \rangle$ represents the expectation operator, $\langle \Theta_{(i,j)} \rangle = \int \Theta_{(i,j)} p(\Theta_{(i,j)}|\mathcal{G}^{(t)})d\Theta_{(i,j)}$, and $p(\Theta_{(i,j)}|\mathcal{G}^{(t)})$ is the posterior distribution which is assumed as Gaussian distribution as follows:

$$p(\Theta_{(i,j)}|\mathcal{G}^{(t)}) = N(W_{(i,j)}, \sigma^2) \tag{7}$$

where $W_{(i,j)}$ is a function of $\mathcal{G}^{(t)}$, which will be discussed later, and then $\langle \Theta_{(i,j)} \rangle = W_{(i,j)}$.

2) M Step: We maximize the ELBO function $Q(\mathcal{G}, \mathcal{G}^t)$ with respect to $\mathcal{G}$, the optimization problem can be reformulated as follows:

$$\begin{aligned} \mathcal{G}^{(t+1)} &= \arg\min_{\mathcal{G}} \; -Q(\mathcal{G}, \mathcal{G}^{(t)}) \\ &= \arg\min_{\mathcal{G}} \; \sum_{i,j,b} W_{(i,j)}|\mathcal{G}_{(i,j,b)}|. \end{aligned} \tag{8}$$

Therefore, according to the optimization objective of the EM algorithm, we propose the definition of TPTV regularizer as follows:

$$\begin{aligned} \|\mathcal{Z}\|_{\text{TPTV}} &= \|\boldsymbol{D}_1\mathcal{Z}\|_{W_1,1} + \|\boldsymbol{D}_2\mathcal{Z}\|_{W_2,1} + \|\boldsymbol{D}_3\mathcal{Z}\|_{W_3,1} \\ &= \sum_{i=1}^{3} \sum_{m,n,b} \left| W_i(m,n)\mathcal{G}_i^{\mathcal{Z}}(m,n,b) \right|. \end{aligned} \tag{9}$$

In this way, using TPTV as a regularization term is essentially maximizing the log-likelihood function $-\sum_{i=1}^{3} \ln(p(\mathcal{G}_i^{\mathcal{Z}}))$.

The determination of weight W_i is the key to the TPTV regularizer. We expect it could relax the sparsity constraints on the pixels with large variations and strengthen the sparsity constraints on the pixels with small variations. Therefore, the weight should have a negative correlation with the image texture and thus can be designed to transform from the texture information. Here, we present a heuristic method to determine the weight. During the denoising processing, we calculate the restoration of HSI data denoted as $\hat{\mathcal{X}}$, but the intensity of residual noise in $\hat{\mathcal{X}}$ is still too strong to degrade the gradient maps. To reduce the adverse effects of residual noise, we perform the low-rank filtering on $\hat{\mathcal{X}}$ to further reduce noise, as follows:

$$\mathcal{X}_0 = \text{LR}(\hat{\mathcal{X}}) \tag{10}$$

where $\text{LR}(\cdot)$ means the low-rank filtering of $\hat{\mathcal{X}}$. In the early stage of denoising, we can set the rank to 1, and thus $\mathcal{X}_0$ only contains the leading component of $\hat{\mathcal{X}}$, and the noise intensity is minimized. With the progress of the denoising process, the image restoration becomes better, and the rank can gradually increase.

After the low-rank filter, the noise intensity is significantly reduced. Although the gradient maps $\mathcal{G}_i^{\mathcal{X}_0}$ can provide us the texture information, the quality is not good enough since the SNR of $\mathcal{G}_i^{\mathcal{X}_0}$ is still low. Fortunately, the image in each band is the imaging results of the same object, so their texture structure is consistent while the noise is random. Therefore, we carry out mean filtering on $\mathcal{G}_i^{\mathcal{X}_0}$ along the spectral dimension to further eliminate the influence of noise. The formula is as follows:

$$G_i = \frac{1}{B}\sum_{b=1}^{B} \left(\mathcal{G}_i^{\mathcal{X}_0}\right)_{(:,:,b)}, \quad i = 1, 2, 3. \tag{11}$$

Note that G_1, G_2, and G_3 are 2-D matrices that provide us with a relatively accurate estimation of texture information along the three modes of HSI.

Then, we transform the texture information into weights. The weights are negatively correlated with the difference value of pixels, so the calculation formula of weights is given as follows:

$$W_i = \frac{\max(|G_i|)}{|G_i| + \delta\max(|G_i|)} \tag{12}$$

where $|\cdot|$ represents the absolution operator, and δ is the threshold adopted to control the range of weight values, usually set to the first quartile of normalized $|G_i|$.

B. ETPTV Denoising Model

The observed HSIs are always contaminated with complex noise. To simplify the model, the noise of HSIs is assumed to be additive, then the observation model of HSIs can be represented as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{E} \tag{13}$$

where $\mathcal{Y}$ is the noisy HSI, $\mathcal{X}$ is the clean HSIs, and $\mathcal{E}$ is the noise term.

We will propose a denoising method based on the TPTV model to recover $\mathcal{X}$ from $\mathcal{Y}$. For consideration of denoising performance, additional HSI priors, such as spectral correlation, need to be introduced into the denoising model. Fortunately, the exploration in ETV [38] confirms that the sparsity of the gradient map can be transformed into a set

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of bases. Thus, we can model each gradient map with the following low-rank decomposition model:

$$\boldsymbol{D}_i(\mathcal{X}) = \mathcal{U}_i \times_3 V_i^T, \quad V_i^T V_i = I \tag{14}$$

where $\mathcal{U}_i \in \mathbb{R}^{M\times N\times r}$ and $V_i \in \mathbb{R}^{B\times r}$ are the representative coefficients and spectral basis, respectively, and r is the rank of $\mathcal{X}$ along the spectral mode. The orthogonal transformation makes the representative coefficients $\mathcal{U}_i$ possibly sufficiently and diversely preserve the knowledge underlying its original gradient map $\boldsymbol{D}_i\mathcal{X}$. Consequently, the weighted l_1 norm can be applied to representative coefficients $\mathcal{U}_i$.

Besides, noise in HSIs is usually modeled as Gaussian noise plus sparse noise. Considering that the TV regularizer has a certain effect on suppressing Gaussian noise [41], we only use the sparse noise to model HSI noise for the purpose of simplifying the denoising model. Thus, our denoising optimization problem is as follows:

$$\begin{aligned}\min_{\mathcal{X},\mathcal{S},U_i,V_i} \ & \|\mathcal{S}\|_1 + \lambda\sum_{i=1}^{3}\|\mathcal{U}_i\|_{W_i,1} \\ \text{s.t. } & \mathcal{Y} = \mathcal{X} + \mathcal{S} \\ & \boldsymbol{D}_i(\mathcal{X}) = \mathcal{U}_i \times_3 V_i^T \\ & V_i^T V_i = I, \quad i = 1,2,3\end{aligned} \tag{15}$$

where the parameter λ is the tradeoff parameter between the TPTV regularization and the noise level.

The proposed denoising method is equivalent to the ETV method [38] if replacing the TPTV regularizer with a traditional 3DTV regularizer. And thus we call the proposed denoising method as ETPTV.

C. Algorithm

The ADMM is used to solve the proposed HSI denoising model. According to the ADMM methodology, the optimization problem described in (15) can be solved by minimizing the following augmented Lagrangian function:

$$\begin{aligned}&L(\mathcal{X},\mathcal{S},\mathcal{U}_i,V_i,\mathcal{M}_i) \\ &= \|\mathcal{S}\|_1 + \lambda\sum_{i=1}^{3}\|\mathcal{U}_i\|_{W_i,1} \\ &\quad + \mu\left\|\mathcal{Y}-\mathcal{X}-\mathcal{S}+\frac{\mathcal{M}_1}{\mu}\right\|_F^2 \\ &\quad + \mu\sum_{i=1}^{3}\left\|\boldsymbol{D}_i(\mathcal{X}) - \mathcal{U}_i\times_3 V_i^T + \frac{\mathcal{M}_{i+1}}{\mu}\right\|_F^2\end{aligned} \tag{16}$$

where $\mathcal{M}_i$, $i = 1,\ldots,4$ are the Lagrange multipliers, and μ is the penalty parameter. Then, each variable can be alternatively updated while keeping all other variables fixed.

1) Update $\mathcal{U}_i$ and V_i: The optimization of $\mathcal{U}_i$ and V_i is performed in two steps. First, we update the spectral subspace basis V_i and representative coefficients $\mathcal{U}_i$ by the following low-rank matrix decomposition problem:

$$\min_{U_i,V_i}\left\|D_i + \frac{1}{\mu}M_{i+1} - U_i V_i^T\right\|_F^2 \tag{17}$$

where D_i, U_i, and M_i represent the unfold of $\boldsymbol{D}_i(\mathcal{X})$, $\mathcal{U}_i$, and $\mathcal{M}_i$ along spectral mode, respectively. This subproblem can be solved by the singular value decomposition (SVD) as follows:

$$\left\{U_i^{\left(t+\frac{1}{2}\right)}, V_i^{(t+1)}\right\} = \text{svd}\left(D_i + \frac{1}{\mu}M_i\right). \tag{18}$$

Then, we project $U_i^{(t+(1/2))}$ into the sparse space as

$$\min_U \ \left\|U_i^{\left(t+\frac{1}{2}\right)} - U\right\|_F^2 + \frac{\lambda}{\mu}\|U_i\|_{W_i,1}. \tag{19}$$

This optimization problem has a closed-form solution, which is represented by the following *soft-thresholding* operator [51]:

$$U_i^{(t+1)} = \boldsymbol{R}_{\frac{\lambda W_i}{\mu}}\left(U_i^{\left(t+\frac{1}{2}\right)}\right). \tag{20}$$

2) Update $\mathcal{S}$: The subproblem with respect to $\mathcal{S}$ can be expressed as

$$\min_{\mathcal{S}}\left\|\mathcal{Y}-\mathcal{X}+\frac{1}{\mu}\mathcal{M}_1-\mathcal{S}\right\|_F^2 + \frac{1}{\mu}\|\mathcal{S}\|_1. \tag{21}$$

Similarly, the solution can be computed by the following *soft-thresholding* operator:

$$\mathcal{S}^{(t+1)} = \boldsymbol{R}_{\frac{1}{\mu}}\left(\mathcal{Y}-\mathcal{X}+\frac{1}{\mu}\mathcal{M}_1\right). \tag{22}$$

3) Update $\mathcal{X}$: The subproblem with respect to $\mathcal{X}$ can be expressed as

$$\begin{aligned}\min_{\mathcal{X}} \ & \left\|\mathcal{Y}-\mathcal{S}+\frac{\mathcal{M}_1}{\mu}-\mathcal{X}\right\|_F^2 \\ & + \sum_{i=1}^{3}\left\|\mathcal{U}_i\times_3 V_i^T - \frac{\mathcal{M}_{i+1}}{\mu} - \boldsymbol{D}_i(\mathcal{X})\right\|_F^2.\end{aligned} \tag{23}$$

This optimization problem can be solved by the following linear system:

$$\begin{aligned}&\left(I + \sum_{i=1}^{3}\boldsymbol{D}_i^*\boldsymbol{D}_i\right)\mathcal{X} = \mathcal{Y}-\mathcal{S}+\frac{\mathcal{M}_1}{\mu} \\ &\qquad + \sum_{i=1}^{3}\boldsymbol{D}_i^*\left(\mathcal{U}_i\times_3 V_i^T - \frac{\mathcal{M}_{i+1}}{\mu}\right)\end{aligned} \tag{24}$$

where $\boldsymbol{D}_i^*$ represents the adjoint operator of $\boldsymbol{D}_i$. The fast Fourier transform can be used to solve this problem, and the solution of $\mathcal{C}$ can be calculated by the following formulas:

$$\begin{cases}\mathcal{H} = \mathcal{Y}-\mathcal{S}+\dfrac{\mathcal{M}_1}{\mu}+\displaystyle\sum_{i=1}^{3}\boldsymbol{D}_i^*\left(\mathcal{U}_i\times_3 V_i - \dfrac{\mathcal{M}_{i+1}}{\mu}\right) \\ \mathcal{T} = |\text{fftn}(\boldsymbol{D}_1)|^2 + |\text{fftn}(\boldsymbol{D}_2)|^2 + |\text{fftn}(\boldsymbol{D}_3)|^2 \\ \mathcal{X}^{t+1} = \text{ifftn}\left(\dfrac{\text{fftn}(\mathcal{H})}{\mathbf{1}+\mathcal{T}}\right)\end{cases} \tag{25}$$

where fftn is the 3-D Fourier transform, ifftn is the inverse transform of fftn, and $\mathbf{1}$ is a tensor with the same size as $\mathcal{T}$ and all of the elements are 1.

After updating the restored HSI $\mathcal{X}$, the weight parameters W_1, W_2, and W_3 can be refreshed as follows:

$$\begin{aligned}\mathcal{G}_i &= \boldsymbol{D}_i(\text{LR}(\mathcal{X}^{t+1})) \\ W_i &= \frac{\max(|\langle\mathcal{G}_i\rangle|)}{|\langle\mathcal{G}_i\rangle| + \delta\max(|\langle\mathcal{G}_i\rangle|)}, \quad i = 1,2,3\end{aligned} \tag{26}$$

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Algorithm 1 ADMM Algorithm for ETPTV

Input: Noisy HSI $\mathcal{Y}$, parameters λ, and threshold δ.
Output: Recovered HSI $\mathcal{X}$.
Initialization: $\mathcal{X} = \mathcal{Y}$, $\mathcal{M}_i$, $i = 1, \ldots, 4$ are zeros tensor. $t = 1$.
1: **while** not coverged **do**
2: Updating $\mathcal{U}_i$ and V_i by (18)-(20).
3: Updating $\mathcal{S}$ by (22).
4: Updating $\mathcal{X}$ by (25).
5: Updating W_i, $i = 1, 2, 3$ by (26).
6: Updating Lagrange multipliers $\mathcal{M}_i$, $i = 1, \ldots, 4$ by (27).
7: $t = t + 1$.
8: **end while**

where $\langle\cdot\rangle$, $\mathrm{LR}(\cdot)$, and δ represent the mean filter along spectral mode, low-rank filter, and the threshold, respectively. The details can be seen in (10)–(12).

The multipliers are updated as follows:

$$\begin{cases} \mathcal{M}_1^{(t+1)} = \mathcal{M}_1^{(t)} + \mu(\mathcal{Y} - \mathcal{X} - \mathcal{S}) \\ \mathcal{M}_i^{(t+1)} = \mathcal{M}_i^{(t)} + \mu\left(\boldsymbol{D}_i\mathcal{X} - \mathcal{U}_i \times_3 V_i^T\right), \quad i = 2, 3, 4. \end{cases} \tag{27}$$

Finally, Algorithm 1 summarizes the whole ADMM algorithm process of solving the ETPTV denoising model.

The algorithm convergence conditions are: 1) the difference between $\mathcal{X}^{(t)}$ and $\mathcal{X}^{(t-1)}$ is less than a threshold value, which is usually set as $1e^{-10}$ and 2) the number of iterations exceeds the presetting maximum number of iterations. The algorithm will terminate when one of the two conditions is satisfied.

D. Computational Complexity Analysis

The computation complexity of ETPTV denoising algorithm mainly includes three parts: 1) updating $\mathcal{U}_i$ and V_i needs the SVD of a matrix with size $(MN) \times B$, and thus the total computational complexity is $O((MN)^2B)$ since $MN \gg B$ for HSI data; 2) updating $\mathcal{X}$, the most time-consuming calculation is the 3-D fast Fourier transform on tensors with size $M \times N \times B$, the complexity of which is $O(MNB \log(MNB))$; and 3) updating weights W_i, the most time-consuming calculation is the SVD of a matrix with size $(MN) \times B$. Hence, the complexity is $O((MN)^2B)$. In summary, the overall computational complexity of our algorithm is $O(K[(MN)^2B + MNB\log(MNB)])$, where K is the number of iterations.

E. Convergence Analysis

All the subproblems involved in the ETPTV algorithm have closed-form solutions. Thus, we have Lemma 1 that the variables $\{\mathcal{M}_i\}_{i=1}^4$ are bounded.

Lemma 1: The sequence of dual variables $\{\mathcal{M}_i\}_{i=1}^4$ generated by Algorithm 1 are bounded.

Then, we can prove that the sequence $(\mathcal{S}^k, \mathcal{X}^k, \mathcal{U}_i^k, V_i^k)$ is a feasible solution of the optimization problem (15), as shown in Lemma 2.

Lemma 2: The accumulation point $(\mathcal{S}^k, \mathcal{X}^k, \mathcal{U}_i^k, V_i^k)$ generated by Algorithm 1 is a feasible solution of the ETPTV model (15).

Based on Lemmas 1 and 2, we can give the following convergence result of Algorithm 1.

Theorem 1: The accumulation point $(\mathcal{S}^k, \mathcal{X}^k, \mathcal{W}_i^k, \mathcal{U}_i^k, V_i^k)$ generated by Algorithm 1 converges to a feasible solution of ETPTV model (15). Suppose the limit point of the sequence $\{\mathcal{S}^k, \mathcal{X}^k, \mathcal{W}_i^k, \mathcal{U}_i^k, V_i^k\}$ is $(\mathcal{S}^*, \mathcal{X}^*, \mathcal{W}_i^*, \mathcal{U}_i^*, V_i^*)$, then $(\mathcal{S}^*, \mathcal{X}^*, \mathcal{W}_i^*, \mathcal{U}_i^*, V_i^*)$ satisfies the Karush–Kuhn–Tucker (KKT) condition of the ETPTV model (15).

The detailed proof of Lemmas 1 and 2 and Theorem 1 are placed in the supplementary material.

V. Experiments

In this section, we implement a series of experiments on synthetic and real HSI datasets to validate the performance of the proposed ETPTV denoising method and the TPTV regularizer. We consider the following HSI denoising methods with representative TV regularizers as comparison methods: the LRTV method [40] using the traditional 2-D TV regularizer band by band, the LRTDTV method [41] adopting the SSTV regularizer on the HSI tensor, the LRTDGS method [47] using the group sparsity TV regularizer, ETV [38] and CTV [38] considering the spectral correlation and sparsity prior on gradient maps simultaneously, RCTV [34] using the 2-D TV regularizer on representative coefficients of HSI data. Moreover, two representative HSI denoising methods are also considered, namely BM4D [29] representing the group filtering method and NGmeet [52] considering multiple image priors simultaneously. In addition, three deep-learning methods HSIDCNN [15], SQAD [18], and SST [21] are used as baseline comparisons with deep-learning methods, which were training on the ICVL hyperspectral dataset. The deep-learning-based methods were run with PyTorch on a machine with Tesla V100 PCIe GPU, Intel(R) Xeon(R) CPU E5-2690 v4 of 2.60 GHz and 503 GB RAM. Other methods were implemented in MATLAB R2021a on a PC with a 3.4 GHz CPU and 32 GB RAM.

A. Simulation Experiments

We perform simulated experiment on two hyperspectral datasets: the CAVE dataset[1] with the size of $512 \times 512 \times 31$ and the Washington DC Mall (DC Mall) dataset[2] whose size is $1208 \times 307 \times 191$. For experimental efficiency, we resize the CAVE dataset to $200 \times 200 \times 31$, and crop the main part of dc Mall to $200 \times 200 \times 191$. The pixel values of the two datasets are normalized to [0, 1]. The two datasets can be used as ground-truth hyperspectral data since they do not contain significant noise visually. To simulate the observed HSI data in real scenarios, we generate the following five types of noise and add them to the ground-truth data.

1) *i.i.d. Gaussian Noise:* Each pixel was contaminated by independent identically distributed (i.i.d.) Gaussian noise with the variance $\sigma = 0.1$.

[1] http://www.cs.columbia.edu/CAVE/databases/
[2] http://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

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TABLE I
PERFORMANCE COMPARISON OF ALL COMPETING METHODS ON THE CAVE DATASET

	Noisy HSI	HSIDCNN	SQAD	SST	BM4D	NGmeet	LRTV	LRTDTV	LRTDGS	CTV	RCTV	ETV	ETPTV
i.i.d. Gaussian Noise													
MPSNR	20.17	35.49	37.33	37.34	38.12	**39.39**	33.40	34.69	35.00	32.95	35.55	34.90	35.63
MSSIM	0.22	0.96	0.97	**0.98**	0.95	0.96	0.90	0.88	0.93	0.83	0.89	0.93	0.93
ERGAS	33.95	5.94	4.74	5.02	4.24	**3.67**	7.23	6.46	6.15	7.85	5.72	6.86	6.31
SAM	0.93	0.23	0.26	0.18	0.21	**0.12**	0.37	0.36	0.27	0.39	0.30	0.28	0.28
Time	-	0.23	0.03	0.18	47.22	209.66	41.14	117.15	21.77	17.96	2.78	11.01	28.05
Non i.i.d. Gaussian Noise													
MPSNR	24.09	37.19	36.86	36.96	**38.64**	37.82	35.69	36.37	36.24	35.00	37.26	36.44	38.19
MSSIM	0.38	0.97	0.96	**0.98**	0.92	0.95	0.93	0.91	0.93	0.88	0.92	0.95	0.95
ERGAS	26.13	4.84	5.21	5.63	4.99	**4.58**	6.70	5.62	5.61	6.18	4.85	5.89	4.94
SAM	0.88	0.21	0.28	0.22	0.26	**0.18**	0.32	0.31	0.23	0.32	0.25	0.23	0.20
Time	-	0.09	0.02	9.19	31.80	116.36	21.17	66.43	21.33	11.84	2.74	7.08	16.20
Gaussian + stripe Noise													
MPSNR	23.39	36.87	36.71	36.70	35.96	36.02	35.07	36.14	35.53	34.63	36.66	36.37	**37.94**
MSSIM	0.37	0.97	0.96	**0.97**	0.91	0.93	0.92	0.91	0.93	0.87	0.91	0.95	0.95
ERGAS	27.07	5.00	5.26	5.76	7.64	6.10	6.92	5.61	6.09	6.39	5.33	5.90	**4.94**
SAM	0.89	0.21	0.28	0.22	0.27	0.23	0.34	0.30	0.27	0.34	0.28	0.222	**0.20**
Time	-	0.09	0.02	9.47	47.04	224.64	25.52	86.34	20.54	13.66	2.77	11.36	21.74
Gaussian + Impulse noise													
MPSNR	20.74	35.57	31.93	36.34	32.11	29.82	33.21	35.14	34.95	34.74	36.33	36.15	**37.75**
MSSIM	0.31	0.94	0.84	**0.97**	0.77	0.79	0.88	0.89	0.95	0.88	0.91	0.95	0.95
ERGAS	62.29	5.89	10.78	6.01	26.56	22.68	16.47	8.29	6.67	6.32	5.51	6.02	**5.09**
SAM	0.97	0.28	0.54	0.25	0.62	0.64	0.46	0.38	0.24	0.30	0.26	0.23	**0.20**
Time	-	0.09	0.02	7.44	44.14	229.95	28.13	104.11	18.16	21.49	2.72	9.92	21.88
Mixture Noise													
MPSNR	21.65	36.04	34.12	36.24	33.26	32.03	34.52	35.46	35.17	34.84	36.33	36.28	**37.92**
MSSIM	0.33	0.95	0.90	**0.97**	0.82	0.85	0.91	0.90	0.94	0.89	0.91	0.95	0.95
ERGAS	48.87	5.53	8.63	6.00	20.99	15.86	11.54	7.50	7.59	6.25	5.54	6.00	**5.02**
SAM	0.94	0.26	0.45	0.25	0.51	0.56	0.38	0.36	0.30	0.31	0.27	0.23	**0.21**
Time	-	0.09	0.02	7.45	48.94	204.96	33.28	118.29	17.85	22.86	2.73	10.56	26.72

2) *Non-i.i.d. Gaussian Noise:* Pixels in different bands were contaminated by i.i.d. Gaussian noise with different variances, and the SNR of each band is randomly distributed in [1, 15] dB.
3) *Gaussian + Stripe Noise:* Each pixel was contaminated by non-i.i.d. Gaussian. Moreover, 20% (for the CAVE dataset) or 30% (for the Washington DC Mall dataset) of bands were randomly chosen to add stripe noise with the stripe number in [10, 30]. In addition, 10% of bands were randomly chosen for adding a deadline with the deadline number in [5, 25].
4) *Gaussian + Impulse Noise:* Each pixel was contaminated by non-i.i.d. Gaussian noise, and 20% of bands were randomly selected to add salt and pepper noise with different degrees.
5) *Mixture Noise:* All pixels were randomly contaminated by at least one type of noise described in case (2)–(4).

To eliminate contingency in experimental results, we repeat the experiment 20 times for each noise case. Four quality assessments were utilized for evaluation, that is, the mean peak SNR (MPSNR) [53] and the mean structural similarity index (MSSIM) [54], the Erreur Relative Globale Adimensionnellede Synthese (ERGAS), and the spectral angle mapper (SAM). Moreover, the experimental time cost of each method is reported to show the efficiency of the method.

The average results of 20 repetitive experiments on the CAVE dataset were shown in Table I. It can be observed that the three deep-learning-based methods HSIDCNN, SQAD, and SST do well in all noise cases, mainly due to that the training dataset ICVL has strong similarity with the CAVE dataset. In the Gaussian noise case, BM4D and NGmeet perform best, while in more complicated noise cases, their performance diminishes gradually since they utilize an excellent image prior model and a rather sample noise model which results in insufficient robustness. Our ETPTV method performs best in complicated noise cases and is nearly the second best in two Gaussian noise cases. In particular, the performance of ETPTV shows significant advantages over ETV, verifying the superiority of the proposed TPTV regularizer. Moreover, the time cost also indicates the proposed method is efficient.

Table II demonstrates the average results on the Washington DC Mall dataset. As can be seen, our method performs best almost in all the cases, and the time cost is also comparable, which states our method is both effective and efficient. Moreover, the ETPTV method substantially outperforms ETV, which further demonstrates the superiority of the TPTV regularizer in encoding the local smoothness prior of HSIs. In addition, the three deep-learning methods perform weakly on this dataset since the discrepancy between test data and training data.

Figs. 2 and 3 show the specific example of denoising results implemented on the two datasets, respectively. For a better visual presentation, a common region in each figure is enlarged in the red box. From these figures, we can see that all methods can remove the mixture noise more or less and that the ETPTV method achieves a remarkable denoising performance and preserve image textures to the best, compared to other methods.

B. Transferring to Other TV Regularizers

In this section, we explore whether the proposed weighting scheme in the TPTV regularizer can be finely embedded

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TABLE II
PERFORMANCE COMPARISON OF ALL COMPETING METHODS ON WASHINGTON DC MALL DATA

	Noisy HSI	HSIDCNN	SQAD	SST	BM4D	NGmeet	LRTV	LRTDTV	LRTDGS	CTV	RCTV	ETV	ETPTV
i.i.d. Gaussian Noise													
MPSNR	20.17	32.23	31.56	33.59	32.51	34.88	34.52	34.23	34.90	34.52	35.44	33.55	**35.51**
MSSIM	0.52	0.96	0.96	**0.97**	0.93	0.96	0.96	0.96	0.96	0.96	0.97	0.95	0.97
ERGAS	23.08	5.65	6.20	5.23	5.35	4.06	4.24	4.36	4.09	4.32	**3.82**	5.02	3.99
SAM	0.47	0.11	0.12	0.11	0.11	**0.07**	0.08	0.09	0.10	0.09	0.08	0.08	0.08
Time	-	0.67	0.35	28.53	255.07	143.43	113.61	330.24	74.11	288.71	7.45	38.24	116.09
Non i.i.d. Gaussian Noise													
MPSNR	20.38	31.46	29.28	33.32	29.95	31.74	32.53	33.76	34.08	34.87	34.77	33.59	**35.80**
MSSIM	0.53	0.95	0.94	0.97	0.87	0.92	0.95	0.95	0.95	0.96	0.95	0.95	**0.97**
ERGAS	26.99	6.06	8.12	6.41	9.25	8.50	8.43	5.75	5.85	4.31	5.43	5.22	**4.00**
SAM	0.59	0.12	0.18	0.12	0.25	0.23	0.23	0.15	0.16	0.10	0.15	0.09	**0.08**
Time	-	0.67	0.35	28.71	249.82	150.98	110.62	341.32	72.43	146.65	7.43	40.14	130.64
Gaussian + stripe Noise													
MPSNR	19.76	31.18	29.17	33.03	29.00	31.66	32.60	33.58	33.61	34.40	34.23	33.31	**35.45**
MSSIM	0.51	0.95	0.93	0.97	0.86	0.92	0.95	0.95	0.95	0.96	0.95	0.95	**0.97**
ERGAS	28.04	6.25	8.22	6.47	9.92	8.54	8.25	5.10	6.52	4.54	5.80	5.35	**4.15**
SAM	0.60	0.12	0.19	0.13	0.25	0.24	0.22	0.11	0.19	0.10	0.16	0.09	**0.09**
Time	-	0.68	0.35	29.25	227.20	178.69	118.05	391.77	71.04	123.85	7.34	46.62	116.77
Gaussian + Impulse noise													
MPSNR	18.45	30.47	28.25	32.79	27.73	29.48	32.66	33.32	33.41	34.56	34.48	33.07	**35.47**
MSSIM	0.46	0.94	0.91	0.97	0.81	0.89	0.95	0.941	0.95	0.96	0.95	0.95	**0.97**
ERGAS	39.21	6.76	9.36	6.82	11.18	12.16	8.24	6.01	6.84	4.47	5.69	5.54	**4.16**
SAM	0.67	0.13	0.22	0.13	0.25	0.25	0.22	0.16	0.19	0.10	0.15	0.09	**0.08**
Time	-	0.65	0.35	28.39	224.34	170.57	117.72	402.11	64.91	123.37	7.20	46.69	125.95
Mixture Noise													
MPSNR	17.95	30.26	28.07	32.51	27.25	29.28	32.41	33.21	33.20	34.18	34.11	33.01	**35.20**
MSSIM	0.44	0.94	0.91	0.96	0.81	0.88	0.95	0.94	0.94	0.96	0.95	0.95	**0.97**
ERGAS	39.64	6.91	9.52	6.86	11.32	12.65	8.32	6.04	6.92	4.67	5.85	5.56	**4.28**
SAM	0.68	0.13	0.22	0.13	0.25	0.26	0.22	0.15	0.19	0.10	0.15	0.09	**0.09**
Time	-	0.69	0.35	30.08	236.76	161.24	103.34	427.21	66.35	161.22	7.1622	43.1524	153.64

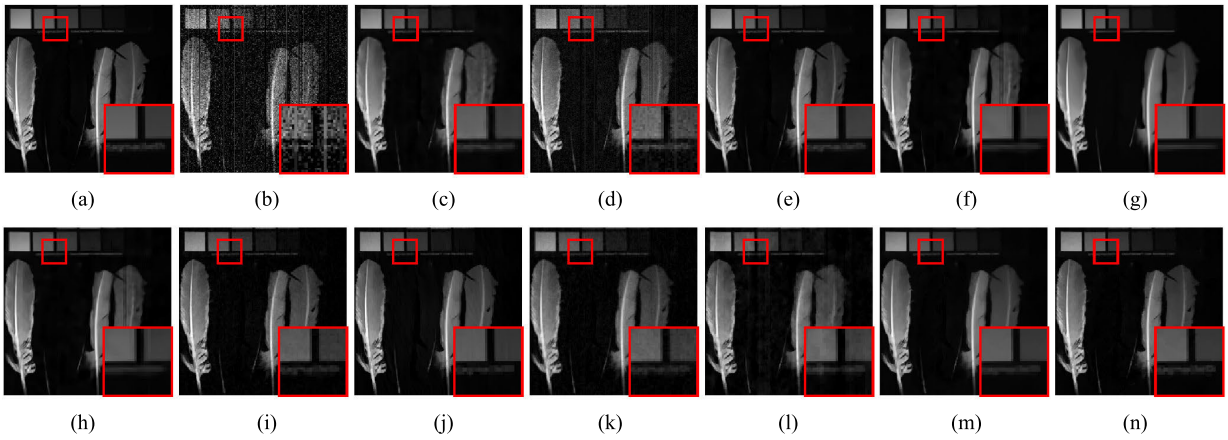


Fig. 2. Image of band 23 in one of the CAVE datasets before and after denoising by different methods. (a) Original data, (b) noisy data, the restored results by method, (c) HSIDCNN, (d) SQAD, (e) SST, (f) BM4D, (g) NGmeet, (h) LRTV, (i) LRTDTV, (j) LRTDGS, (k) CTV, (l) RCTV, (m) ETV, and (n) ETPTV.

into other TV regularizers. To this aim, we choose three HSI denoising methods adopted by different TV regularizers. Specifically, the LRTV method [40] utilized the traditional 2-D TV regularizer band by band, the LRTDTV method [41] considered the distribution of pixel variations along three modes and set different weights for each gradient map, and the RCTV method [34] adopted the 2-D TV regularizer on representative coefficients of HSI data. We extend these methods by embedding the texture-preserved weights as described in (3)–(12) into their TV regularizer, namely LRTPTV, LRTDTPTV, and RCTPTV. No extra hyperparameters are introduced to the original models, and the optimization algorithm is only needed to change the part referring to TV regularization.

We conduct the comparison experiment on the CAVE dataset with the same noise settings as Section V-A, and the quantitative results are shown in Table III. As can be seen, the denoising performance has been greatly promoted after embedding texture-preserved weights in their TV regularizer, which indicates that the weighting scheme can be finely transferred to other TV regularizers and significantly improve the performance of the original model.

Fig. 4 shows two instances of denoising results under mixture noise obtained by original methods and its corresponding weighted vision. It can be observed that all the methods can remove noise well, but the method after embedding texture-preserved weights has clearer texture information and achieves better visual results.

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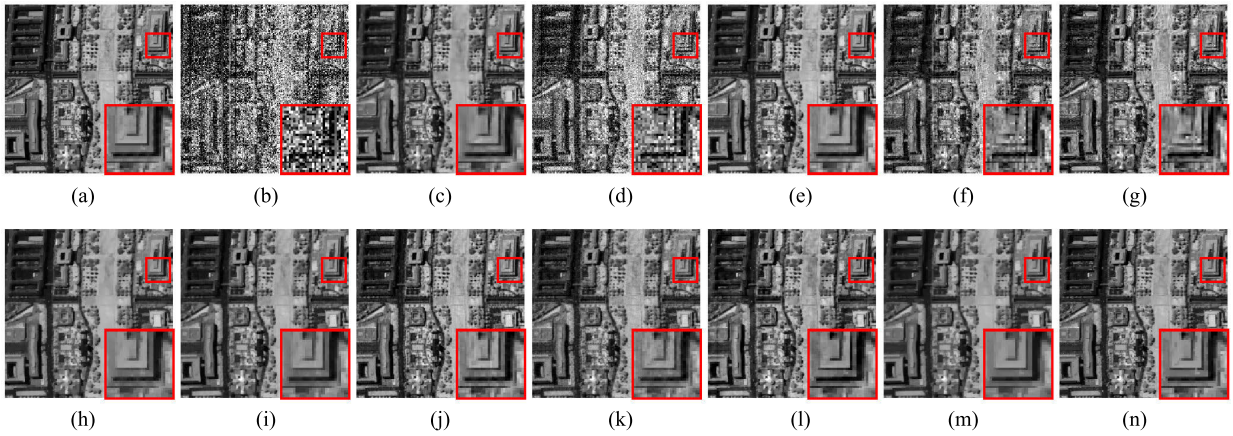


Fig. 3. Image of band 69 in the Washington DC Mall dataset before and after denoising by different methods. (a) Original data, (b) noisy data, the restored results by method, (c) HSIDCNN, (d) SQAD, (e) SST, (f) BM4D, (g) NGmeet, (h) LRTV, (i) LRTDTV, (j) LRTDGS, (k) CTV, (l) RCTV, (m) ETV, and (n) ETPTV.

TABLE III
PERFORMANCE COMPARISON OF ALL COMPETING METHODS ON THE CAVE DATASET

Noise Types	Metric	Noisy HSI	LRTV	LRTPTV	LRTDTV	LRTD-TPTV	RCTV	RCTPTV
i.i.d. Gaussian Noise	MPSNR	20.172	33.995	**34.845**	34.688	**35.577**	35.547	**36.856**
	MSSIM	0.215	0.897	**0.911**	0.875	**0.921**	0.887	**0.938**
	ERGAS	33.945	7.227	**6.332**	6.457	**5.938**	5.720	**4.883**
	SAM	0.932	0.372	**0.296**	0.363	**0.290**	0.295	**0.181**
Non i.i.d. Gaussian Noise	MPSNR	24.092	35.691	**35.764**	36.374	**37.309**	37.262	**38.135**
	MSSIM	0.384	**0.925**	0.924	0.909	**0.942**	0.919	**0.950**
	ERGAS	26.127	6.702	**6.204**	5.621	**5.042**	4.849	**4.391**
	SAM	0.884	0.324	**0.304**	0.314	**0.223**	0.249	**0.161**
Gaussian + Stripe Noise	MPSNR	23.391	35.069	**35.308**	36.142	**37.154**	36.659	**37.566**
	MSSIM	0.367	0.915	**0.918**	0.907	**0.941**	0.905	**0.939**
	ERGAS	27.074	6.924	**6.418**	5.605	**5.043**	5.333	**4.752**
	SAM	0.891	0.341	**0.306**	0.303	**0.225**	0.278	**0.194**
Gaussian + Impulse Noise	MPSNR	20.739	33.208	**33.880**	35.139	**35.750**	36.330	**37.326**
	MSSIM	0.305	0.883	**0.889**	0.891	**0.920**	0.910	**0.940**
	ERGAS	62.287	16.470	**11.886**	8.290	**7.833**	5.506	**4.836**
	SAM	0.970	0.459	**0.436**	0.375	**0.317**	0.257	**0.182**
Mixture Noise	MPSNR	21.646	34.520	**34.854**	35.455	**36.345**	36.331	**37.359**
	MSSIM	0.325	0.906	**0.907**	0.899	**0.932**	0.905	**0.937**
	ERGAS	48.868	11.536	**8.902**	7.496	**6.623**	5.537	**4.850**
	SAM	0.937	0.379	**0.353**	0.358	**0.282**	0.273	**0.195**

C. Real Data Experiments

This section evaluates the performance of our method on real scenarios by conducting the experiments on real HSI datasets, that is, Urban dataset,[3] from which a portion of data with size 256 × 256 × 210 was cropped for the experiment, and Indian Pines[4] whose size is 145 × 145 × 220. The two datasets are seriously polluted.

1) Urban Dataset: Figs. 5 and 6 show the denoising results of the 108 and 207 bands in the Urban dataset, respectively. As can be observed, the deep-learning-based methods HSID-CNN, SQAD, and SST can remove noise significantly, but lead to image distortion since the biased deep priors. The recovered results of BM4D and NGmeet contain obvious noise residue due to the serious discrepancy between simplistic i.i.d. Gaussian noise assumption and the real situation. Although LRTV and LRTDTV suppress noise obviously, they also result in oversmooth issues. Among all these methods, our approach obtains the best performance both in noise removal and texture preservation.

To further evaluate each method, the spectral signatures of the pixel with spatial coordinate (185, 132) were drawn in Fig. 7. The horizontal and vertical axes in the figure represent the band number and digital number value of the given location, respectively. The spectral signature curves of the original HSI fluctuate rapidly because of the presence of noise, as shown in Fig. 7(a). All denoising methods suppress these fluctuations more or less, among which ETV and ETPTV suppress the fluctuations most significantly, and it is noteworthy that ETPTV is capable of keeping small changes in the spectral curve. In summary, our ETPTV method performs best both in noise removal and texture preservation.

2) AVIRIS Indian Pines Dataset: Figs. 8 and 9 show the recovered results of bands 103 and 220, respectively.

[3]http://www.tec.army.mil/hypercube
[4]https://engineering.purdue.edu/~biehl/MultiSpec/hyperspectral.html

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Fig. 4. Image of one band in the CAVE dataset before and after denoising by different methods and its weighted version. (a) Original data, (b) noisy data, the restored results by method, (c) LRTV, (d) LRTPTV, (e) LRTDTV, (f) LRTDTPTV, (g) RCTV, and (h) RCTPTV.

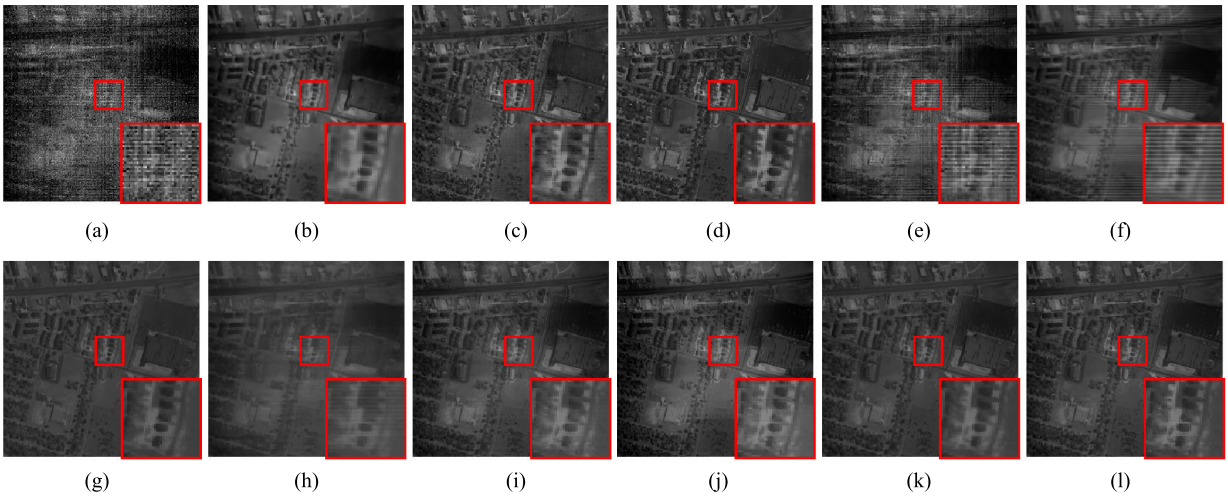


Fig. 5. Image of band 108 in the Urban dataset before and after denoising by different methods. (a) Original data, the restored results by method, (b) HSIDCNN, (c) SQAD, (d) SST, (e) BM4D, (f) NGmeet, (g) LRTV, (h) LRTDTV, (i) CTV, (j) RCTV, (k) ETV, and (l) ETPTV.

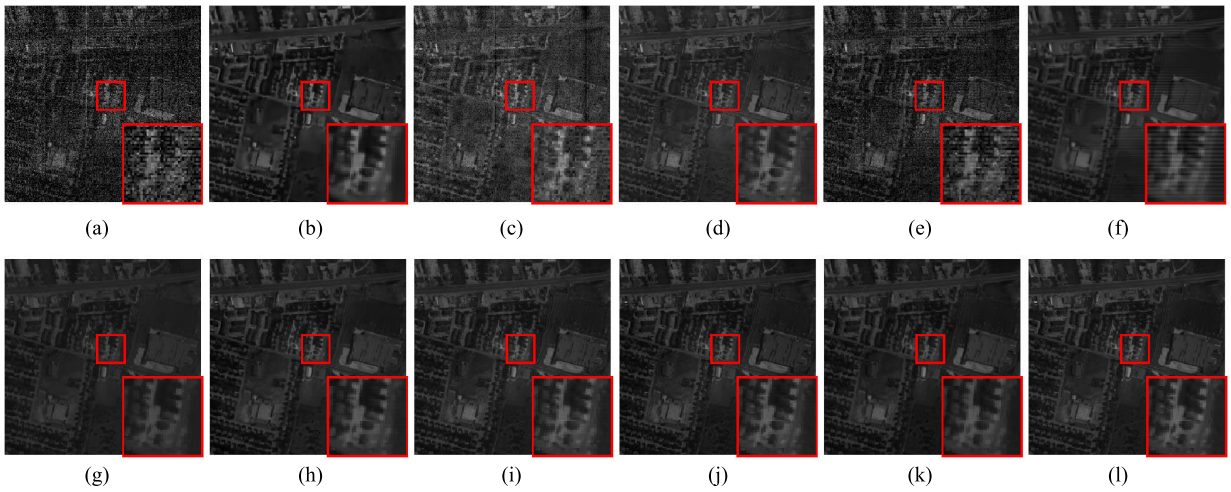


Fig. 6. Image of band 207 in the Urban dataset before and after denoising by different methods. (a) Original data, the restored results by method, (b) HSIDCNN, (c) SQAD, (d) SST, (e) BM4D, (f) NGmeet, (g) LRTV, (h) LRTDTV, (i) CTV, (j) RCTV, (k) ETV, and (l) ETPTV.

It can be seen that BM4D loses its effectiveness in the case of heavy mixture noise. Although the comparison methods can remove the noise to some extent, some noise or visual image distortion remains in the restored image. ETV and ETPTV obtain the best visual denoising results. In addition, the spectral signatures of the pixel with spatial coordinate $(125, 20)$ were drawn in Fig. 10, where rapid fluctuations of the curves can be observed, especially in heavily contaminated bands as shown in Fig. 10(a). After denoising by methods BM4D, LRTV, and NGmeet, the curves become much more smooth, but the obvious fluctuations can still be observed in these bands, indicating that the noise is

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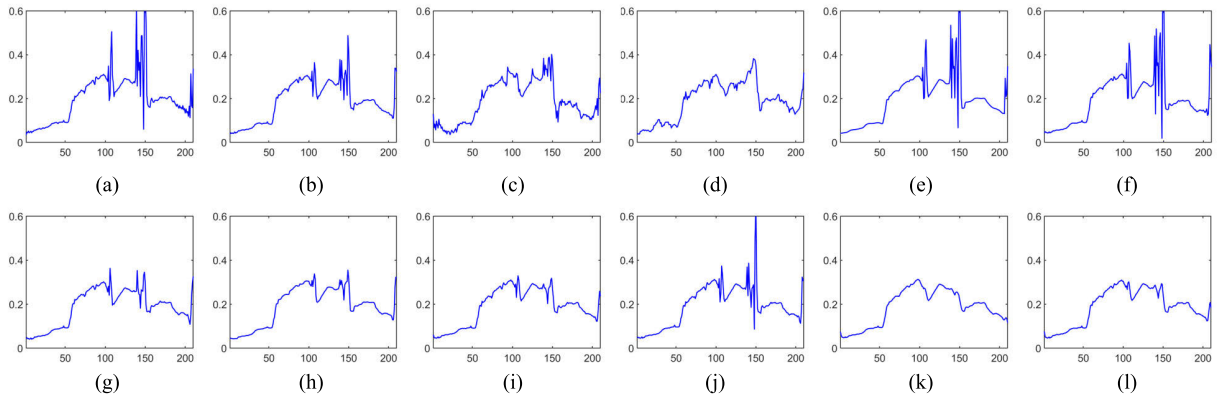


Fig. 7. Spectral signatures of point (185, 132) in the Urban dataset before and after denoising by different methods. (a) Original data, the restored results by method, (b) HSIDCNN, (c) SQAD, (d) SST, (e) BM4D, (f) NGmeet, (g) LRTV, (h) LRTDTV, (i) CTV, (j) RCTV, (k) ETV, and (l) ETPTV.

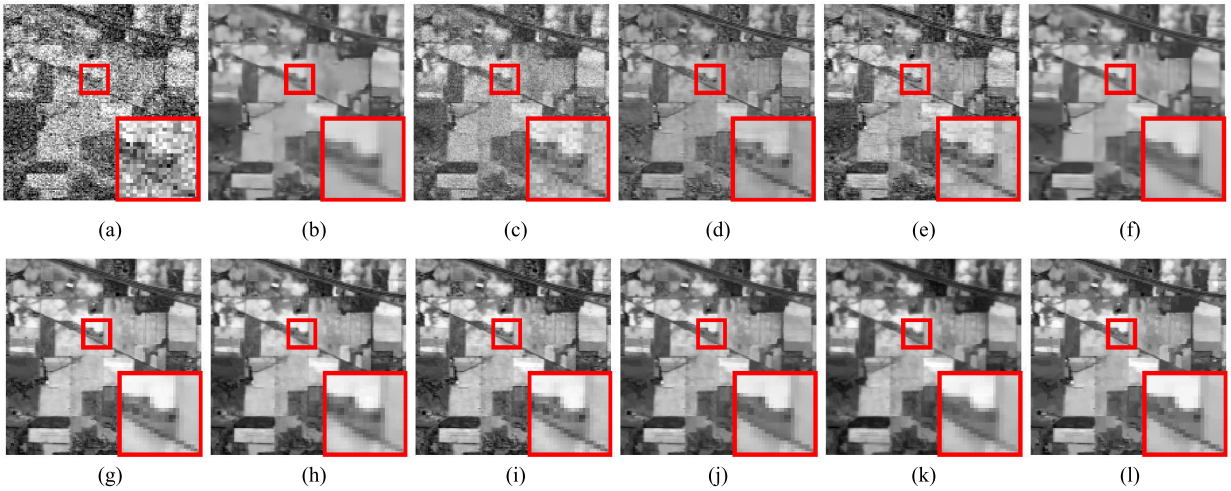


Fig. 8. Image of band 103 in the Indian Pines dataset before and after denoising by different methods. (a) Original data, the restored results by method, (b) HSIDCNN, (c) SQAD, (d) SST, (e) BM4D, (f) NGmeet, (g) LRTV, (h) LRTDTV, (i) CTV, (j) RCTV, (k) ETV, and (l) ETPTV.

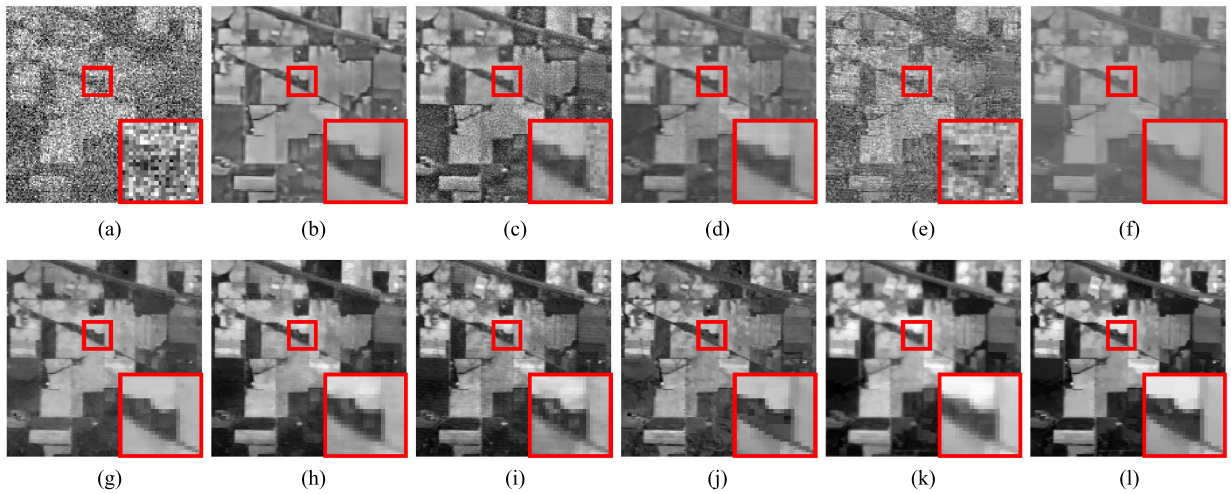


Fig. 9. Image of band 220 in the Indian Pines dataset before and after denoising by different methods. (a) Original data, the restored results by method, (b) HSIDCNN, (c) SQAD, (d) SST, (e) BM4D, (f) NGmeet, (g) LRTV, (h) LRTDTV, (i) CTV, (j) RCTV, (k) ETV, and (l) ETPTV.

still present. The curves of other methods are more stable, which indicates that the noise is well suppressed on all the bands.

To further study the influence of denoising processing for downstream applications, a classification experiment on the Indian Pines dataset was conducted. The utilized classification method is the support vector machine (SVM). The data can be divided into 16 categories, with the number of samples in each category ranging from 20 to 2455. We randomly select 10 samples from each category for training and the rest for testing. Three quantitative criteria [55], namely overall accuracy (OA), average accuracy (AA), and the kappa coefficient (kappa), are

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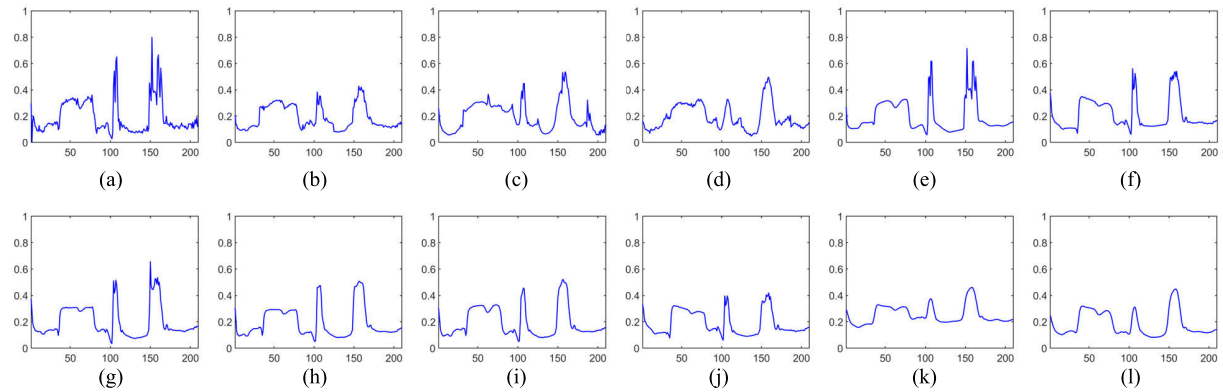


Fig. 10. Spectral signatures of point (125, 20) in the Indian Pines dataset before and after denoising by different methods. (a) Original data, the restored results by method, (b) HSIDCNN, (c) SQAD, (d) SST, (e) BM4D, (f) NGmeet, (g) LRTV, (h) LRTDTV, (i) CTV, (j) RCTV, (k) ETV, and (l) ETPTV.

TABLE IV
CLASSIFICATION ACCURACY COMPARISON ON THE AVIRIS INDIAN PINES DATA

	Noisy HSI	HSIDCNN	SQAD	SST	BM4D	NGmeet	LRTV	LRTDTV	CTV	RCTV	ETV	ETPTV
OA	0.536	0.744	0.528	0.722	0.610	0.726	0.700	0.710	0.708	0.709	0.742	**0.754**
AA	0.620	**0.839**	0.645	0.800	0.711	0.809	0.801	0.818	0.806	0.814	0.826	0.830
kappa	0.476	0.713	0.475	0.684	0.561	0.690	0.664	0.676	0.673	0.673	0.709	**0.722**

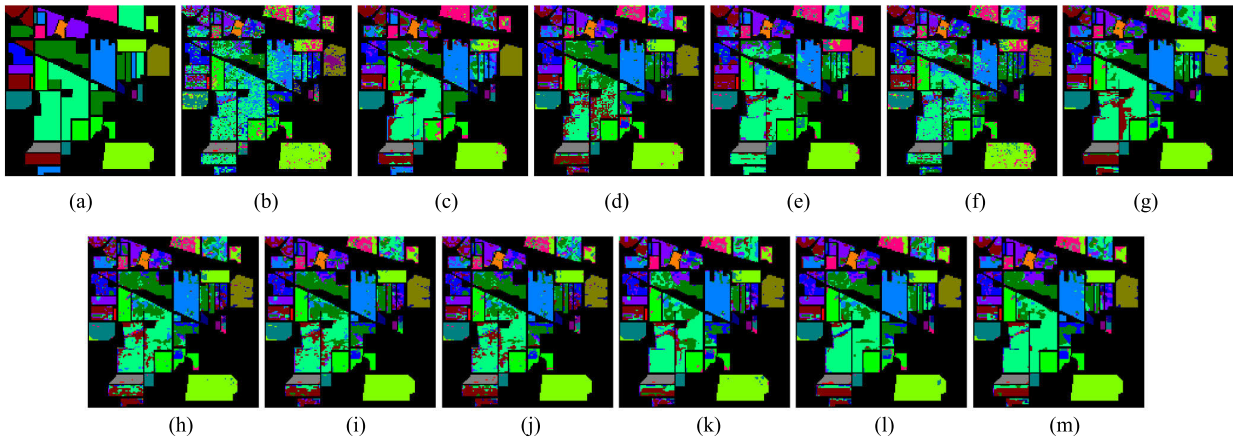


Fig. 11. Classification results by performing SVM on the Indian Pines dataset before and after denoising, and the corresponding OA values are reported. (a) Ground-truth classification, (b) classification result on original data, and the classification results on restored data by methods, (c) HSIDCNN, (d) SQAD, (e) SST, (f) BM4D, (g) NGmeet, (h) LRTV, (i) LRTDTV, (j) CTV, (k) RCTV, (l) ETV, and (m) ETPTV.

utilized to evaluate the classification results. The classification experiment is repeated ten times, that is, training and testing on ten random training and testing datasets. The best results obtained from these experiments are presented in Table IV, and the corresponding label maps are depicted in Fig. 11. As can be seen from Table IV and Fig. 11, the classification accuracy of the original HSI is rather low due to the presence of noise. After denoising by all the compared methods, the accuracy is improved more or less. Among all the denoising methods, the proposed method exhibits the highest OA and kappa values, indicating a significant contribution of its denoising results to the subsequent tasks.

D. Parameter Analysis

The tradeoff parameter λ and ranks $r_i (i = 1, 2, 3)$ are model parameters that need to be preset. In all experiments, we set the ranks of the gradient maps along the two spatial modes as r_1, and the rank of the gradient map along the spectral mode as r_2. To analyze the impact of parameters λ, r_1, and r_2 on denoising performance, we conduct denoising experiments on one of the CAVE datasets under a set of parameter settings for both stripe noise and mixture noise. The denoising performance was measured with MPSNR. The MPSNR change curve versus $\lambda \in [5e^{-4}, 1.5e^{-2}]$ with $r_1 = 7, r_2 = 5$ is depicted in Fig. 12(a), where the curves demonstrate a consistent pattern for both types of noise. Specifically, when λ is small, the TPTV regularization carries less weight in the model, resulting in a higher presence of noise in the denoised image. Consequently, the MPSNR values are significantly low. As λ gradually increases, more noise is eliminated, causing MPSNR to improve until reaching the peak. Further increasing the value of λ results in excessive intensity of regularization, leading to the loss of original image information. Consequently, the values of MPSNR gradually decrease. Nevertheless, when λ is within a certain range, MPSNR consistently remains at a high level, indicating an acceptable sensitivity of this parameter in practical applications. Furthermore, λ controls the intensity of TPTV regularization. Its value

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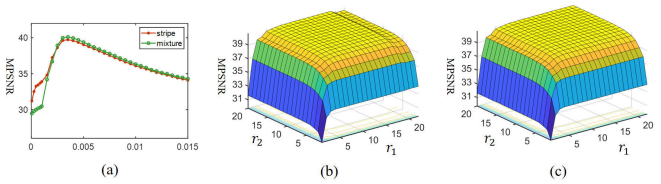


Fig. 12. (a) MPSNR change versus parameter λ with $r_1 = 7$ and $r_2 = 5$. MPSNR change versus r_1 and r_2 with $\lambda = 3.5\text{e}^{-4}$ in (b) stripe noise case and (c) mixture noise case.

can be determined based on the smoothness of the image. A smoother image would require a larger value of λ, whereas a less smooth image would necessitate a smaller value of λ.

The MPSNR change versus $r_1 \in [1, 22]$ and $r_2 \in [1, 20]$ while keeping $\lambda = 3.5\text{e}^{-4}$ for stripe noise and mixture noise cases are depicted in Fig. 12(b) and (c), respectively. The MPSNR change demonstrates a consistent pattern for both types of noise. Specifically, when r_1 and r_2 are too small, the original image information will be lost, resulting in significantly low values of MPSNR. As $r_1 \geq 5$ and $r_2 \geq 5$ rising, MPSNR increases rapidly and consistently remains at a high level, indicating that the rank selection is robust.

VI. Conclusion

This article proposes a novel TPTV regularizer with weighted l_1 norm of gradient maps based on the LSM distribution model. The weights are adaptively determined from the denoised image, allowing for a reduction in the sparsity constraint on pixels with large variations. This adaptive weighting helps preserve image texture and prevents oversmoothing. Based on the TPTV regularizer, a new denoising model named ETPTV is developed by adopting the weighted l_1 norm on representative coefficients. Experimental results illustrate that our method outperforms state-of-the-art methods, highlighting the TPTV regularizer's ability to capture the nonlocal smoothness properties of images better than other TV regularizers. In addition, the proposed weighting scheme can be well embedded into other existing TV regularizers and help to protect the image texture information from being smoothed. The experiments also prove that after embedding this weighting scheme, the denoising performance of the original methods is significantly promoted.

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